

Structured summary: Critical evaluation of the modified Gompertz response function as a parametric template for treatment-response trajectories in N-of-1 trial simulation

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This document is a structured, section-by-section summary of the key points argued in the companion manuscript, “Critical evaluation of the modified Gompertz response function as a parametric template for treatment-response trajectories in N-of-1 trial simulation.” It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations,

34 **0.1 A claim about response-curve choice in N-of-1 trial simulation**

35 **The Gompertz function is a defensible default for the saturating-shape regime in**
36 **N-of-1 simulation.**

- 37 • Its asymmetric three-parameter form spans profiles that logistic and exponential families
38 cannot represent.
- 39 • Established precedent in tumor dynamics, microbial growth, and the unified-Richards
40 family supports the choice.
- 41 • The claim is conditional on the saturating-shape regime characteristic of most pharma-
42 codynamic trajectories over typical trial timescales.

43 **Family-misspecification sensitivity is absent from the N-of-1 literature and should**
44 **be reported routinely.**

- 45 • Existing N-of-1 simulation studies inherit trajectory families without comparing alter-
46 natives.
- 47 • Oncology and pharmacology modeling have established comparative apparatus that the
48 N-of-1 field has not imported.
- 49 • Routine sensitivity reporting is needed rather than silent assumption of family adequacy.

50 **The case for Gompertz is incomplete without a direct comparison of inferential**
51 **consequences across alternative families.**

- 52 • The paper does not assert Gompertz is uniquely correct but argues it is reasonable
53 within the unified-Richards class.
- 54 • Inferential consequences (power, Type I error, bias) under alternative families must be
55 quantified, not asserted.
- 56 • The simulation study uses the standard pmsimstats linear-mixed analysis at trial-
57 relevant sample sizes.

58 **0.2 Why the question matters**

59 **Family choice is a DGP-level decision whose inferential cost cannot be detected**
60 **without varying the DGP family while holding the analysis fixed.**

- 61 • A single-family simulation cannot reveal the inferential cost of that family choice.
- 62 • Robustness to family is a joint property of the DGP and analysis specification.
- 63 • This is analysis-model misspecification logic applied one level down in the simulation
64 hierarchy.

65 **The Gompertz parameterization has known identifiability fragility in short noisy**
66 **trajectories that has not been characterized for N-of-1 designs.**

- 67 • Asymptote and rate parameters correlate strongly; the shape parameter is near-
68 unidentifiable with few inflection- region timepoints.
- 69 • Unified-Richards reparameterizations and sloppiness diagnostics exist but have not been
70 applied to N-of-1 trajectory data.
- 71 • Comparative identifiability results from tumor-growth modeling support the need for
72 an analogous N-of-1 diagnosis.

73 **The Gompertz default in N-of-1 simulation reflects historical borrowing from**
74 **growth biology, not empirical comparison with pharmacology alternatives.**

- 75 • The pmsimstats Gompertz form traces to tumor dynamics and microbial growth litera-
76 ture, not pharmacodynamic evidence.
- 77 • The pharmacology literature defaults to Hill/sigmoid-Emax and indirect-response mod-
78 els for comparable questions.
- 79 • The two traditions have not been compared; inheritance is not itself evidence of ade-
80 quacy.

81 **0.3 What the published N-of-1 literature does and does not do**

82 **No published N-of-1 simulation study has examined parametric trajectory-family**
83 **choice or reported family-misspecification sensitivity.**

- 84 • CONSORT-N-of-1, foundational programmatic statements, and the Hendrickson frame-
85 work all omit trajectory-family comparison.
- 86 • Silent inheritance of the Gompertz form is pervasive across applied N-of-1 simulation
87 work.
- 88 • Adjacent literatures (oncology, pharmacometrics, latent- class growth) have comparative
89 apparatus that has not been applied to N-of-1 simulation.

90 **Each major N-of-1 methodological strand silently inherits the Gompertz default**
91 **without family comparison.**

- 92 • CONSORT-N-of-1 reporting standards, Lillie and Schork programmatic statements, and
93 the Hendrickson framework all omit family comparison.
- 94 • Applied prazosin-PTSD work inherits the framework without revisiting the trajectory
95 assumption.
- 96 • No published N-of-1 simulation study, to the authors' knowledge, has broached the
97 misspecification question.

98 **Comparative trajectory-family apparatus exists in adjacent literatures but has**
99 **not been applied to N-of-1 simulation.**

- 100 • Oncology modeling, pharmacometrics, and latent-class growth literature each have es-
101 tablished multi-family comparison methods.

- None of these methods has been applied with the Gompertz form as the N-of-1 simulation reference.
- Importing this machinery is the explicit contribution of the present paper.

The paper's contribution is a four-family comparison against the standard pmsimstats analysis, reported under the ADEMP framework.

- Four families are compared: Gompertz, symmetric logistic, hyperbolic-tangent, and piecewise-linear breakpoint.
- The analysis model is held fixed at the standard pmsimstats linear-mixed specification.
- Results are reported in the Morris-White-Crowther ADEMP framework adopted as the project-wide standard.

0.4 What this paper contributes

Section 2 characterizes the pmsimstats Gompertz parameterization and its relationship to the unified-Richards family.

- The vertical-offset adjustment and calibration parameter sets are documented.
- The unified-Richards reparameterization provides more interpretable, less correlated parameters.
- This section establishes the reference point for all subsequent sensitivity comparisons.

Section 3 surveys three alternative families at matched marginal effect size for visual and structural comparison with Gompertz.

- Families covered: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- Each is calibrated to match the Gompertz reference at the same asymptote and anchor-point response level.
- Shape differences (symmetry, curvature, smoothness) are the primary axes of contrast.

Sections 4 and 5 report a Monte Carlo sensitivity simulation comparing power, Type I error, and bias across the four trajectory families.

- Each family generates trajectories; all are analyzed under the fixed standard pmsimstats linear-mixed model.
- Performance measures are reported with conventional MCSEs.
- The design borrows the Benzekry eight-model comparative methodology and the Jagadeesan sloppiness diagnostic.

Section 6 identifies the trajectory regimes where Gompertz misspecification would be inferentially consequential and recommends sensitivity-grid extensions.

- Consequential regimes include non-saturating growth, biphasic patterns, and breakpoint behavior.
- Standard N-of-1 inference is materially affected by family choice in these regimes.
- Specific sensitivity-grid extensions are recommended for pmsimstats users.

138 **A companion technical document preserves full mathematical and historical con-**
139 **text omitted from the manuscript proper.**

- 140 • Location: docs/25-gompertz-model-evaluation.md.
- 141 • Contains citations, derivations, and historical context that did not survive manuscript
- 142 editing.
- 143 • The manuscript is self-contained; the companion document is supplementary reference
- 144 material.

145 **1 The modified Gompertz function in pmsimstats**

146 **The pmsimstats framework adopts a modified monotone- saturating Gompertz**
147 **function for each of three latent response components.**

- 148 • Provenance traces to Gompertz’s 1825 mortality law and Winsor’s 1932 growth-curve
- 149 application.
- 150 • Three components are parameterized: biological response BR, placebo-belief response
- 151 PB, and natural-history TV.
- 152 • The modified form is monotone-saturating, distinguishing it from the original mortality-
- 153 curve application.

154 **The three-parameter Gompertz form is asymmetric and monotone-saturating,**
155 **with the inflection location determined jointly by rate and displacement.**

- 156 • Parameters are maxr (asymptote), rate (approach speed), and disp (curvature near
- 157 $t = 0$); a vertical offset ensures $y(0) = 0$.
- 158 • Inflection at $t^* = \log(\text{disp})/\text{rate}$; the asymmetry is the key structural distinction from
- 159 logistic.
- 160 • Prazosin/PTSD reference values are $\text{maxr}_{BR} \approx 4$, $\text{rate}_{BR} \approx 0.5$, $\text{disp}_{BR} \approx 4$.

161 **2 Alternative response-curve families**

162 **Three alternative families are calibrated to match Gompertz at a common anchor**
163 **point, isolating shape effects from amplitude effects.**

- 164 • Families compared: symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint.
- 165 • Calibration constraint: matched asymptote and response level at $t = 5$ weeks via uniroot
- 166 or closed form.
- 167 • Families differ in smoothness, inflection symmetry, and curvature while sharing the same
- 168 marginal effect size.

169 **2.1 Symmetric logistic**

170 **The symmetric logistic rises faster than Gompertz before the inflection and slower**
171 **after, with symmetric inflection as the principal structural contrast.**

- Symmetric around inflection at t_0 ; a vertical offset ensures $y(0) = 0$.
- At matched asymptote and anchor, the logistic envelope differs visibly from Gompertz in pre- and post-inflection curvature.
- Inflection symmetry is the primary qualitative contrast with the asymmetric Gompertz.

2.2 Hyperbolic-tangent

The hyperbolic-tangent family represents the fast- saturating smooth regime that neither the logistic nor Gompertz can span with standard parameterizations.

- Symmetric and monotone-saturating, but with a steeper pre-asymptote rise than the logistic.
- Used in pharmacokinetic modeling for drugs with rapid receptor saturation.
- Its inclusion spans the fast-saturating smooth regime absent from the other three families.

2.3 Piecewise-linear breakpoint

The piecewise-linear breakpoint is the canonical non- smooth alternative and provides the strongest test of the framework's sensitivity to trajectory non-smoothness.

- A linear ramp to breakpoint t_{bp} then constant or small post-breakpoint slope.
- The breakpoint is the sole non-differentiable feature across the four families.
- Whether the Wald test detects the non-smooth feature is the central empirical question the simulation addresses.

3 Sensitivity-analysis simulation design

The primary estimand is the `bm:Dbc` interaction coefficient; performance measures are power, Type I error, bias, and empirical MCSE.

- Analysis model: $Sx \sim bm + t + Dbc + bm:Dbc + (1|ptID)$ with `corCAR1(form = ~t|ptID)`.
- Primary outputs: power at $\alpha = 0.05$ and Type I error under $\beta_{bm:D} = 0$.
- Secondary outputs: bias and empirical MCSE of $\hat{\beta}_{bm:D}$; all proportions reported with binomial MCSE.

The ADEMP design is pre-registered and fixes all four DGP families, the calibration sub-study, and the full factorial before production runs.

- Pre-registration location: `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md`.
- Study 0 calibrates all four families to a common marginal effect size; Study 1 is the 144-cell factorial; Study 2 is the sloppiness eigenvalue diagnostic.
- Pre-registration prevents post-hoc specification of performance measures or DGP choices.

207 **The present submission uses a focused 16-cell factorial scoped to answer whether**
208 **trajectory family matters for the `bm:Dbc` Wald test at MCSE 0.013–0.016.**

- 209 • Design: 4 families $\times$ 2 architectures $\times$ 2 effect sizes; prazosin-calibrated OL+BDC,
210 $N = 70$, $t_{1/2} = 0$.
- 211 • Carryover is absent to isolate trajectory shape from carryover-mediated correlation de-
212 cay.
- 213 • The 144-cell extended grid is deferred to a follow-up; the 16-cell factorial addresses the
214 central question at adequate MCSE resolution.

215 **All four families are matched at $t = 5$ weeks with the asymptotic maximum held**
216 **constant; calibrated parameters are recorded for reproducibility.**

- 217 • Anchor calibration: each family’s response at $t = 5$ matched to Gompertz via uniroot
218 or closed form.
- 219 • Asymptotic maximum held constant across all four families.
- 220 • Calibrated parameters recorded in `02-faithful-trajectory-sweep.R` and transcribed
221 into the summary at `07-gompertz-faithful-summary.txt`.

222 **The trajectory family drives the data-generating process directly, and under the**
223 **mean-moderation architecture the biomarker moderation is trajectory-scaled, in**
224 **keeping with the paper-01 multiplicative model.**

- 225 • The family sets the BR mean curve inside the covariance construction (`buildSigma`
226 `br_family`), not by a post-hoc additive shift.
- 227 • Under the mean-moderation architecture the per-timepoint moderation scales with the
228 response trajectory, normalized so the mean on-drug magnitude equals σ_{BR} (matched
229 marginal effect).
- 230 • Under the covariance-moderation architecture the interaction is in the covariance and
231 is mean-independent, so the family enters the BR mean only.

232 4 Results

233 **The 16-cell simulation completed with 100% convergence in 597 s; cell MCSE on**
234 **power runs from 0.013 to 0.014.**

- 235 • 1,000 replicates per cell across 16 cells; 8-core `parallel::mclapply`.
- 236 • Per-replicate output at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`;
237 ADEMP summary at `07-gompertz-faithful-summary.txt`.
- 238 • Results are presented in order: power, Type I error, bias.

239 4.1 Power across response-curve families

240 **The inferential consequence of trajectory family is architecture-dependent: ex-**
241 **actly zero under the covariance channel, modest but non-zero under faithful mean**
242 **moderation.**

- The covariance-moderation architecture: with matched seeds the four families return identical rejection decisions, so family is inert by construction; the production-seed spread is sampling noise.
- The mean-moderation architecture: families span 0.743 to 0.782, a 0.039 spread; Gompertz (the default) is the lowest-power family, roughly two MCSE below logistic.
- The pre-registered hypothesis that the piecewise-linear breakpoint would behave qualitatively differently is not supported; it tracks the logistic, not the Gompertz.

The earlier flat-moderation result, which reported no family effect under either architecture, was an artifact of the interaction encoding, not a property of trajectory family.

- The prototype applied the biomarker moderation as a flat shift, independent of the response trajectory, so family could not affect the interaction under either architecture.
- Once the moderation is trajectory-scaled per the paper-01 Architecture-A model, the mean-moderation families separate.
- The covariance-channel invariance is unchanged, because that channel is genuinely mean-independent.

4.2 Type I error across response-curve families

Type I error is slightly conservative across all four families, consistent with corCAR1 small-sample degree-of-freedom accounting.

- All eight null cells fall in $[0.019, 0.038]$; no family-specific inflation is observed.
- At the null the moderation term vanishes, so the trajectory-scaling of the mean-moderation architecture is inactive and the null cells are governed by the same mechanism as before.
- The conservatism is shared across families, implicating the corCAR1 d.f. mechanism rather than trajectory shape.

4.3 Bias of the biomarker-treatment interaction estimate

Bias of the interaction estimate is negligible and family-invariant under both architectures, so the Architecture-A power separation is a precision effect, not a bias effect.

- Mean $\hat{\beta}_{bm:D_{bc}}$ varies from -0.230 to -0.239 across the eight alternative cells, a 0.009 spread within Monte Carlo noise.
- Empirical SE is essentially constant across families (0.073 to 0.083).
- The matched-anchor calibration holds the estimand fixed across families, so the power spread reflects the temporal profile of the signal rather than its magnitude or bias.

277 4.4 An architecture-dependent finding

278 **The inferential consequence of trajectory family depends on how the biomarker-**
279 **treatment interaction is encoded: zero under the covariance channel, modest**
280 **under mean moderation.**

- 281 • Under the covariance-moderation architecture (covariance channel) the family is inert
282 by construction; the interaction test is mean-independent.
- 283 • Under the mean-moderation architecture the four families separate over a 0.039 power
284 range, with the Gompertz default the lowest-power, that is mildly conservative, family.
- 285 • The effect is modest (about two MCSE at $N = 70$, $c_{bm} = 0.45$) and should not be
286 overstated, but it is real and direction-consistent.

287 **The pre-registered piecewise-linear hypothesis is not supported in the anticipated**
288 **direction; the breakpoint family tracks the smooth alternatives, not the Gompertz**
289 **default.**

- 290 • The non-differentiable breakpoint carries no distinctive power signature at $N = 70$ and
291 $c_{bm} = 0.45$.
- 292 • Under the mean-moderation architecture the breakpoint family sits with the logistic
293 and hyperbolic-tangent, above the Gompertz default.
- 294 • Smoothness per se is not the operative axis; the temporal profile of the moderation is.

295 4.5 Why the architectures differ

296 **Two mechanisms explain the contrast: exact mean- independence of the covari-**
297 **ance channel, and the residual profile-sensitivity of the trajectory-scaled mean**
298 **channel.**

- 299 • Under the covariance-moderation architecture the biomarker-treatment information
300 lives in the covariance and is mathematically independent of the trajectory mean, so
301 the family cannot move the test.
- 302 • Under the mean-moderation architecture the moderation rides the trajectory, so the
303 temporal profile of the signal, not just its anchored magnitude, enters the Wald statistic.
- 304 • Matched-anchor calibration leaves only profile differences, which the time-averaging
305 Wald test damps but does not erase.

306 **Matched-anchor calibration ensures that residual family differences are in shape**
307 **only, not in amplitude, and those shape differences are small relative to between-**
308 **replicate variation.**

- 309 • Families matched at asymptote and $t = 5$ anchor.
- 310 • Within-trajectory shape variation is small relative to the between-replicate variance of
311 the `bm:Dbc` estimator.
- 312 • The calibration design isolates shape effects from amplitude effects by construction.

313 **The `bm:Dbc` Wald test averages over on-drug timepoints, smoothing out local**

314 **trajectory-shape differences that would be visible in point-in-time comparisons.**

- 315 • The continuous D_{bc} predictor integrates trajectory contributions across all on-drug time-
316 points.
- 317 • Local shape differences are averaged within the integration and do not accumulate to
318 detectable power differences.
- 319 • This temporal averaging damps, but does not erase, the profile differences that drive
320 the Architecture-A spread.

321 **Under the covariance-moderation architecture the interaction is in the covariance**
322 **structure and is exactly independent of trajectory shape, which we verify by a**
323 **matched-seed check.**

- 324 • The covariance-moderation architecture encodes information in $\partial\text{Cor}(BM, BR)/\partial D_{bc}$,
325 not in the trajectory mean.
- 326 • Holding the seed fixed, the four families return identical rejection decisions, so the
327 invariance is exact, not merely within Monte Carlo noise.
- 328 • This is the mechanism behind the exact Architecture-B inertness, in contrast to the
329 merely damped Architecture-A effect.

330 **The two mechanisms together place the modified Gompertz default as inferen-**
331 **tially free under the covariance channel and mildly conservative under the mean**
332 **channel.**

- 333 • Under the covariance-moderation architecture the default is robust by exact construc-
334 tion; trajectory family cannot matter.
- 335 • Under the mean-moderation architecture the default is the lowest-power family, so it is
336 a safe, slightly conservative, choice rather than an optimal one.
- 337 • The following section identifies conditions that would amplify the Architecture-A effect.

338 4.6 Conditions that would change the answer

339 **Four conditions outside the present 16-cell factorial would plausibly amplify the**
340 **Architecture-A family effect; none would activate an effect under the mean-**
341 **independent the covariance-moderation architecture.**

- 342 • Smaller N , different c_{bm} , non-zero carryover ($t_{1/2} > 0$), and non-monotone or biphasic
343 shape envelopes.
- 344 • None of these conditions is exercised in the present focused factorial.
- 345 • Each is a natural axis for the 144-cell follow-up grid; the biphasic case is the only one
346 expected to reach the covariance-moderation architecture.

347 **Smaller sample size might amplify local-shape sensitivity by concentrating tem-**
348 **poral information at fewer timepoints.**

- 349 • At $N = 70$, the test integrates over a moderately large within-subject sample.

- With fewer timepoints, local shape differences between families would be more visible rather than averaged.
- This condition is the most proximate candidate for amplifying the Architecture-A effect beyond a few percent.

At smaller c_{bm} near the 50% power boundary, family differences might surface via different finite-sample sampling distributions.

- The $c_{bm} = 0.45$ cell sits in a moderate-power regime with comparable signal-to-noise across families.
- Pushing to $c_{bm} = 0.30$ (power near 0.5) may reveal family-specific finite-sample sampling differences.
- The tight clustering at 0.45 may be a local feature of the effect-size space.

Non-zero carryover introduces an interaction between trajectory shape and the off-drug exponential decay that is suppressed in the present $t_{1/2} = 0$ slice.

- With $t_{1/2} > 0$, integration weights interact with the off-drug decay, potentially differentiating families.
- The present simulation sets $t_{1/2} = 0$ to isolate trajectory shape from carryover effects.
- This interaction is the mechanistic pathway by which carryover could make trajectory family consequential.

Non-monotone or biphasic trajectory families would produce genuinely different `bm:Dbc` information content to which the Wald test would almost certainly respond.

- All four present families share a monotone-saturating envelope; the small-effect finding is conditional on this property.
- Biphasic or oscillating families have qualitatively different information structure.
- Extension to non-monotone families is outside the scope of the present study.

The 144-cell extended grid is the canonical follow-up that would close the trajectory-family question at publication- adequate rigor.

- The grid crosses N , c_{bm} , $t_{1/2}$, and family.
- Each of the four conditions above maps to a dimension of this grid.
- The present focused factorial is a necessary precursor; the extended grid is the production-grade completion.

4.7 The prazosin/PTSD reference cell

For the prazosin-PTSD application, trajectory family is a low-consequence sensitivity axis relative to carryover and DGP-architecture, but not a free one under mean moderation.

- Sensitivity to the modified Gompertz function is small relative to carryover specification and DGP-architecture sensitivity.

- Under a covariance-channel DGP the modified Gompertz choice is inferentially free; under mean moderation it is mildly conservative, costing a few percentage points of power.
- Sensitivity-analysis resources are better spent on carryover and DGP-architecture axes with larger inferential consequences, while still reporting one family comparison under the mean-moderation DGP.

5 Conclusions

The modified Gompertz is a defensible template for monotone-saturating N-of-1 trajectories; the inferential cost of the choice is architecture-dependent.

- Under the covariance channel (the covariance-moderation architecture) the four families are exactly equivalent; under mean moderation (the mean-moderation architecture) they span a 0.039 power range at MCSE 0.013–0.014.
- The Gompertz default is the lowest-power family under mean moderation, that is mildly conservative, not optimal.
- The piecewise-linear breakpoint hypothesis is not supported; the breakpoint family tracks the smooth alternatives.

The finding implies the Wald test is robust to trajectory family under the covariance channel and mildly sensitive to it under mean moderation; boundary conditions are deferred to the 144-cell follow-up.

- Robustness is exact under the covariance-moderation architecture and approximate (few-percent) under the mean-moderation architecture in the evaluated regime.
- At what sample sizes, effect sizes, or carryover levels the Architecture-A effect grows is an open question.
- Sensitivity-analysis priority should go to carryover and DGP-architecture axes with larger demonstrated consequences.

The 144-cell extended grid is the work needed to close the trajectory-family question at publication-adequate rigor.

- The present 16-cell factorial addresses the central question with statistical clarity.
- The extended grid would map boundary conditions at which trajectory family begins to matter.
- This follow-up is recommended as the production-grade completion of the trajectory-family evaluation.

6 Data availability statement

Data and code are openly available in the pmsimstats-ng repository under versioned standard paths.

- Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are at `analysis/data/`.
- Simulation drivers are at `analysis/scripts/`.
- Exact manuscript paths are listed in the Reproducibility section.

7 Reproducibility

The manuscript is fully reproducible inside the zzcollab Docker container with the `renv.lock` package set.

- Container image hash in `Dockerfile`; package set in `renv.lock`.
- Principal packages: `nlme`, `parallel`, `data.table`, `furrr/purrr`, `rmarkdown/bookdown`.
- Rendered with R 4.4.x.

Seeds, checkpoints, and driver paths are fully specified for exact replication of the simulation results.

- Master seed `set.seed(20260617)`; per-cell seed offsets give independent but reproducible replicate streams.
- Outputs checkpointed at `analysis/data/quick-sim/07-gompertz-faithful-replicates.rds`; ADEMP summaries at `07-gompertz-faithful-summary.txt`.
- Driver `02-faithful-trajectory-sweep.R`; pre-registered ADEMP plan at `analysis/scripts/gompertz`